

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Module - I

Practical –V

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5.Analyzing Data using Table Calculations in Tableau.

a) Apply quick table calculations (running total, percent of total).

STEPS:-

- 1.Open a new worksheet and rename it: Running Total – Votes.
2. From the Data Pane, drag listed_in(city) to the Rows shelf.
3. Drag votes to the Columns shelf. (Tableau will automatically aggregate it as SUM(votes)).
4. Convert to Quick Table Calculation:

Right-click the SUM(votes) pill on the Columns shelf.

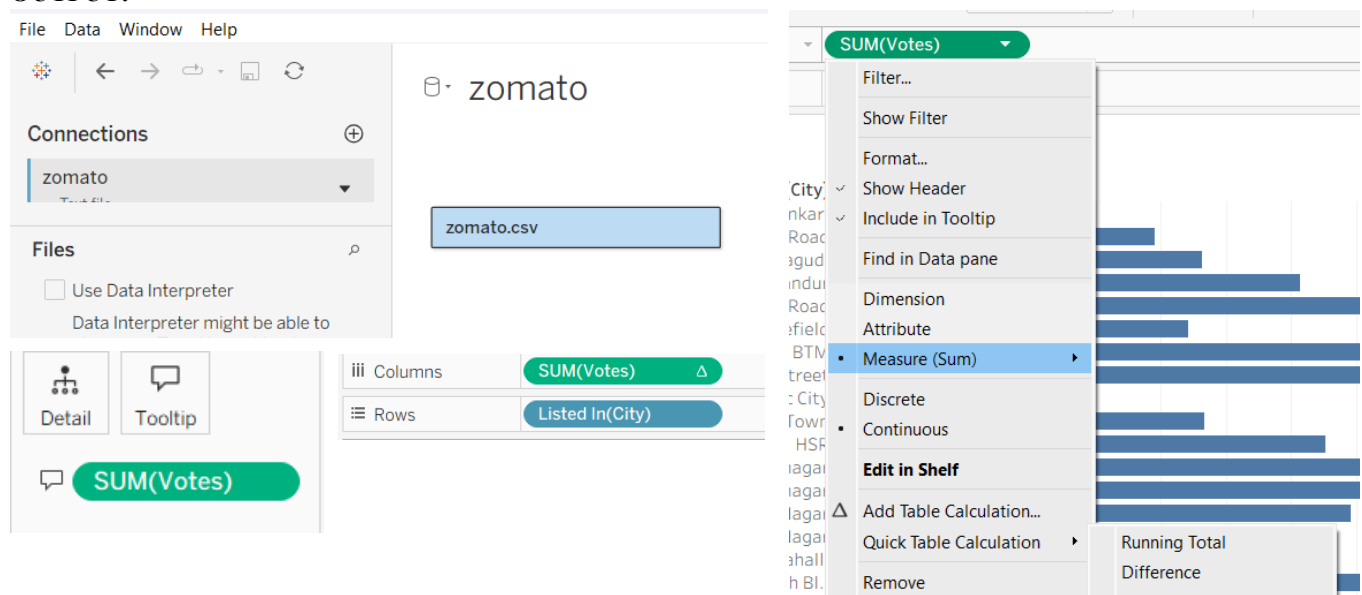
Hover over Quick Table Calculation and select Running Total.

5. Add Visual Context (Secondary Measure):

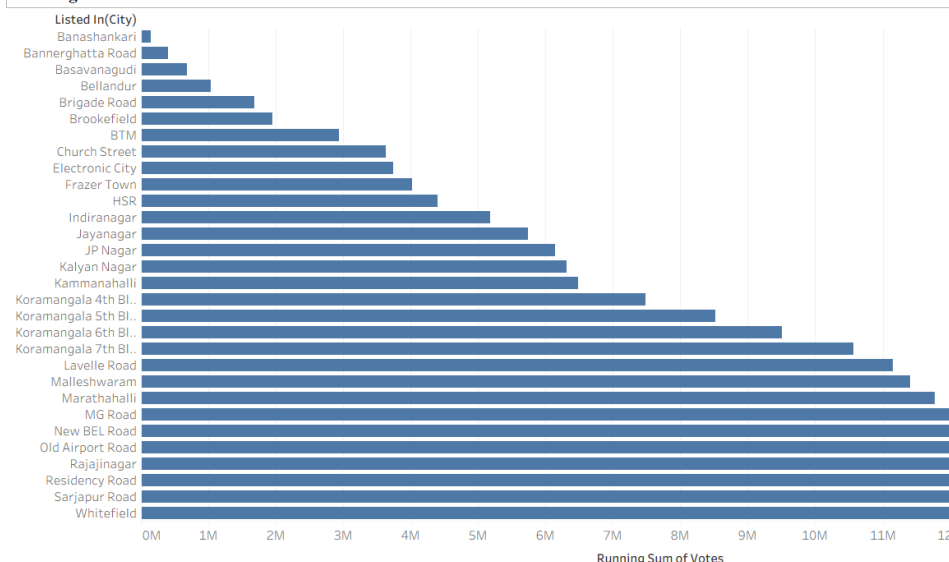
Drag votes from the Data Pane again and drop it directly onto the Tooltip mark in the Marks Card.

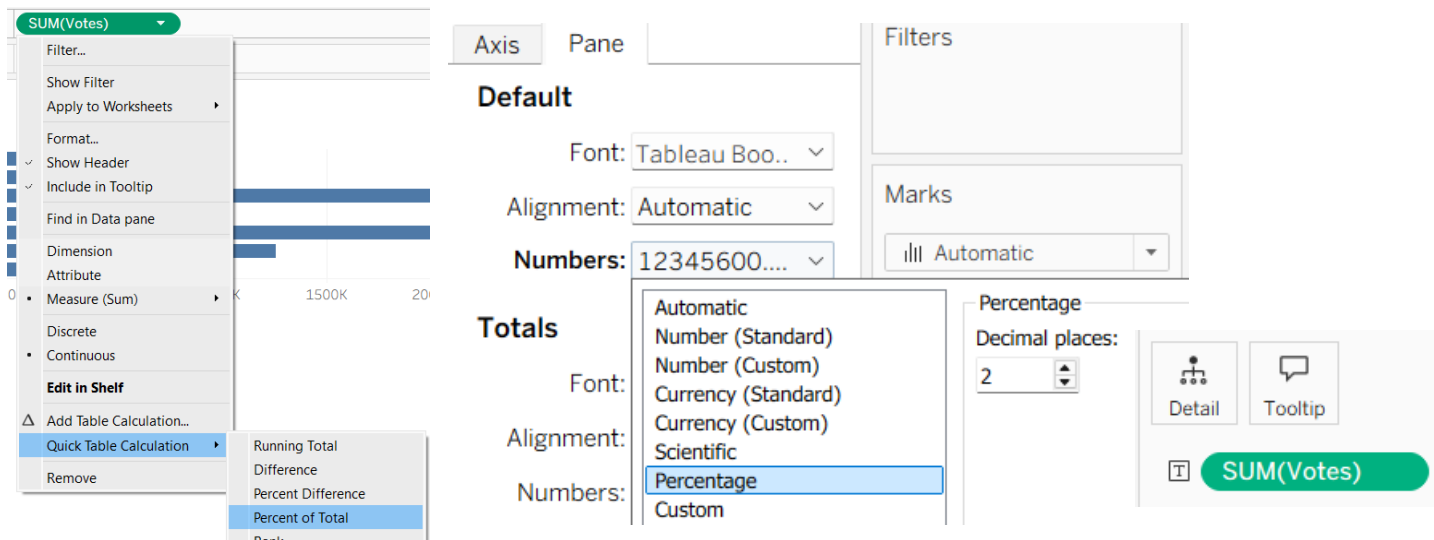
Result: Now when you hover over a bar, you will see both the individual city's vote count and the cumulative total up to that city.

OUTPUT:-

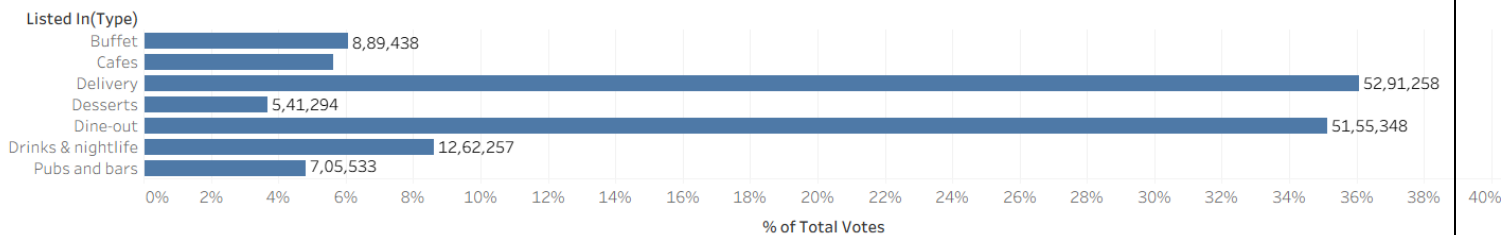


Running total T054





Percent of total T054

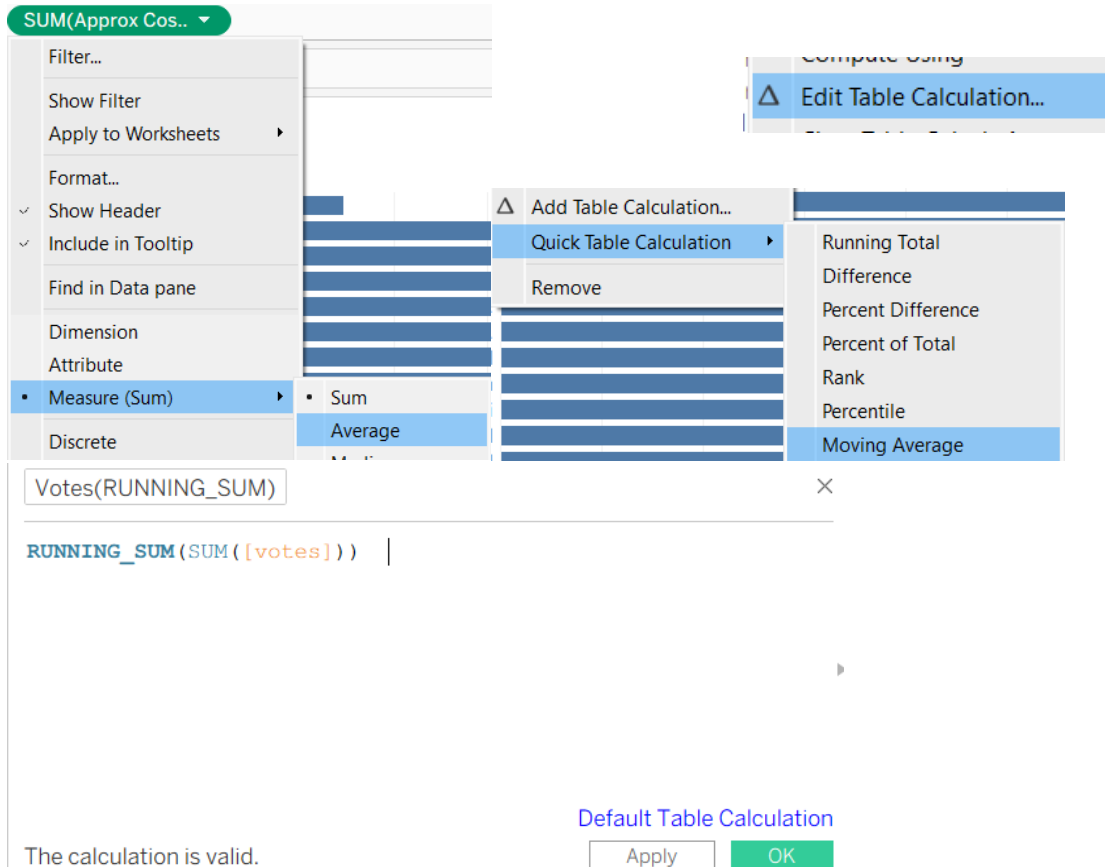


b) Create moving averages and cumulative metrics.

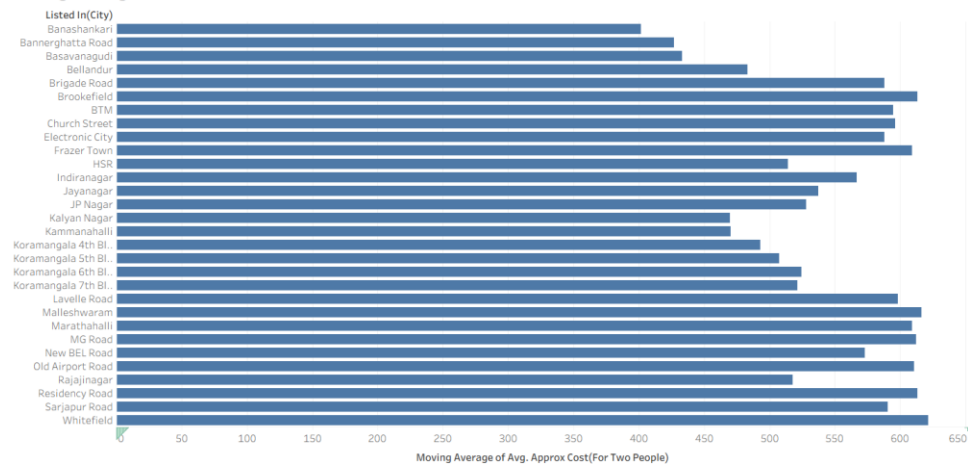
STEPS:-

1. Open a new worksheet and rename it: Moving Average - Avg Cost.
2. Drag listed_in(city) to the Rows shelf.
3. Drag approx_cost(for two people) to the Columns shelf.
4. Convert to a Moving Average Quick Table Calculation:
Right-click the AVG pill on the Columns shelf. Hover over Quick Table Calculation and select Moving Average.
5. Drag cuisines to Rows (Filter to top 15 or 20 if the list is too long).
7. Drag your newly created Cumulative Votes (RUNNING_SUM) field onto Columns.

OUTPUT:-



Moving Average-AVG Cost T054



c) Perform year-over-year growth analysis.

STEPS:-

1. Open a new worksheet and rename it: YoY Growth Analysis.
2. Place your time dimension (or ordinal grouped column like listed_in(city)) on the Columns shelf.
3. Drag votes (or approx_cost(for two people)) onto the Rows shelf as SUM(votes).
4. Apply the Year-over-Year Growth Quick Table Calculation:

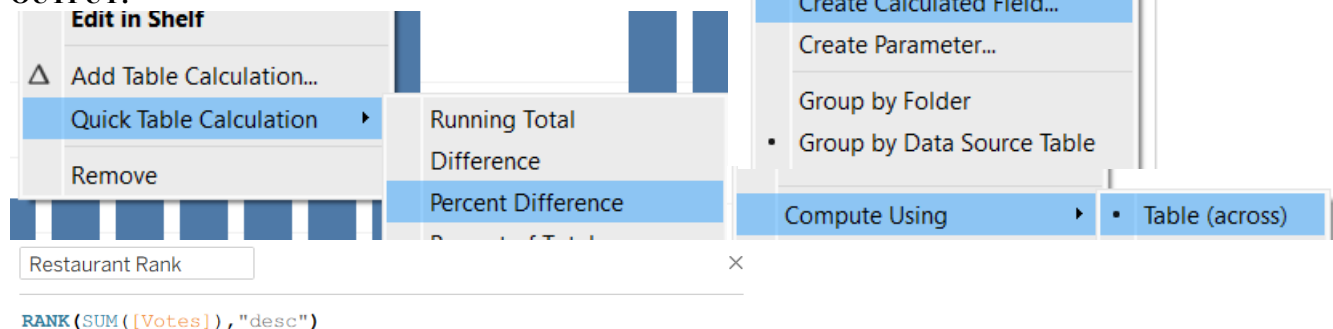
Right-click the SUM(votes) pill on the Rows shelf.

Hover over Quick Table Calculation $\diamond$ select Percent Difference (or Year Over Year Growth if using a true Date field).

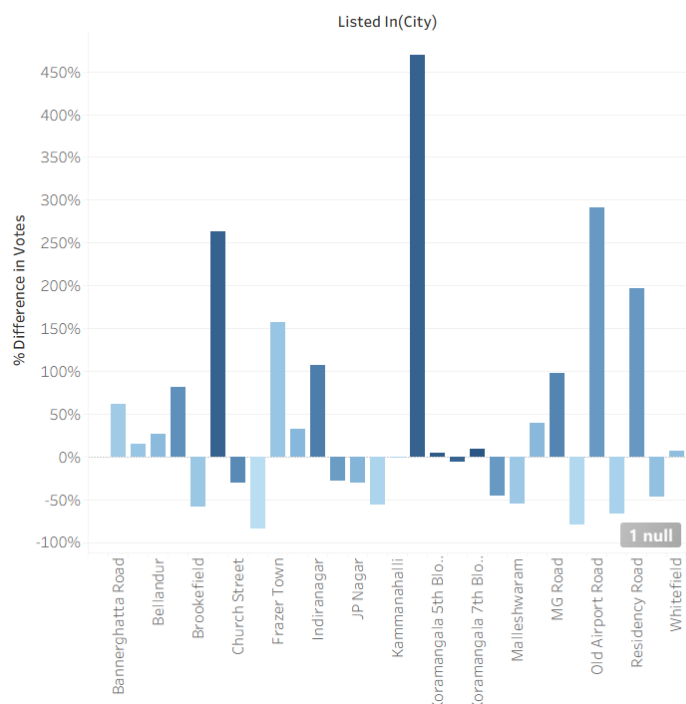
5. Configure the Growth Calculation Direction.

6. Add Visual Color Indicators.

OUTPUT:-



yoy Growth Analysis T054



d) Use ranking functions (Top N, Bottom N).

STEPS:-

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...
2. Name the field: Restaurant Rank.
3. Input the following ranking calculation: `RANK(SUM([votes]), 'desc')`
4. Build the Top N / Bottom N Ranked View
5. Open a new worksheet and rename it: Top & Bottom Restaurants. Drag name (Restaurant Name) to the Rows shelf.
6. Drag votes to the Columns shelf.
7. Drag your newly created Restaurant Rank calculated field onto the Rows shelf, placing it to the left of name.
8. In the filter dialog box, select a range of ranks
9. Click OK.

OUTPUT:-

The screenshot shows the Tableau interface. On the left, the 'Columns' shelf contains 'SUM(Votes)' and the 'Rows' shelf contains 'Name'. A context menu is open for the 'Restaurant Rank' field, showing options like 'Filter...', 'Show Filter', 'Format...', 'Show Header', 'Include in Tooltip', 'Find in Data pane', 'Discrete', and 'Continuous'. The 'Filter [Restaurant Rank]' dialog box is open, showing the 'General' tab. It has radio buttons for 'Select from list' (selected), 'Custom value list', and 'Use all'. Below these is a search bar and a list of values from 1 to 11, with checkboxes next to them. The 'All' button is selected. At the bottom, there are buttons for 'Reset', 'OK', 'Cancel', and 'Apply'.

City Window Avg Cost

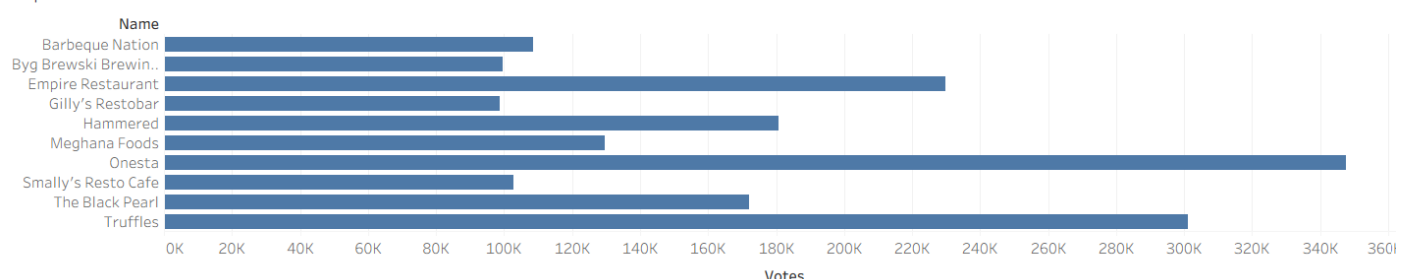
`WINDOW_AVG(AVG([Approx Cost (For Two People)]))`
`WINDOW_AVG(expression, [start, end])`

Default Table Calculation

The calculation is valid.

Apply OK

top and Bottom Restaurants T054

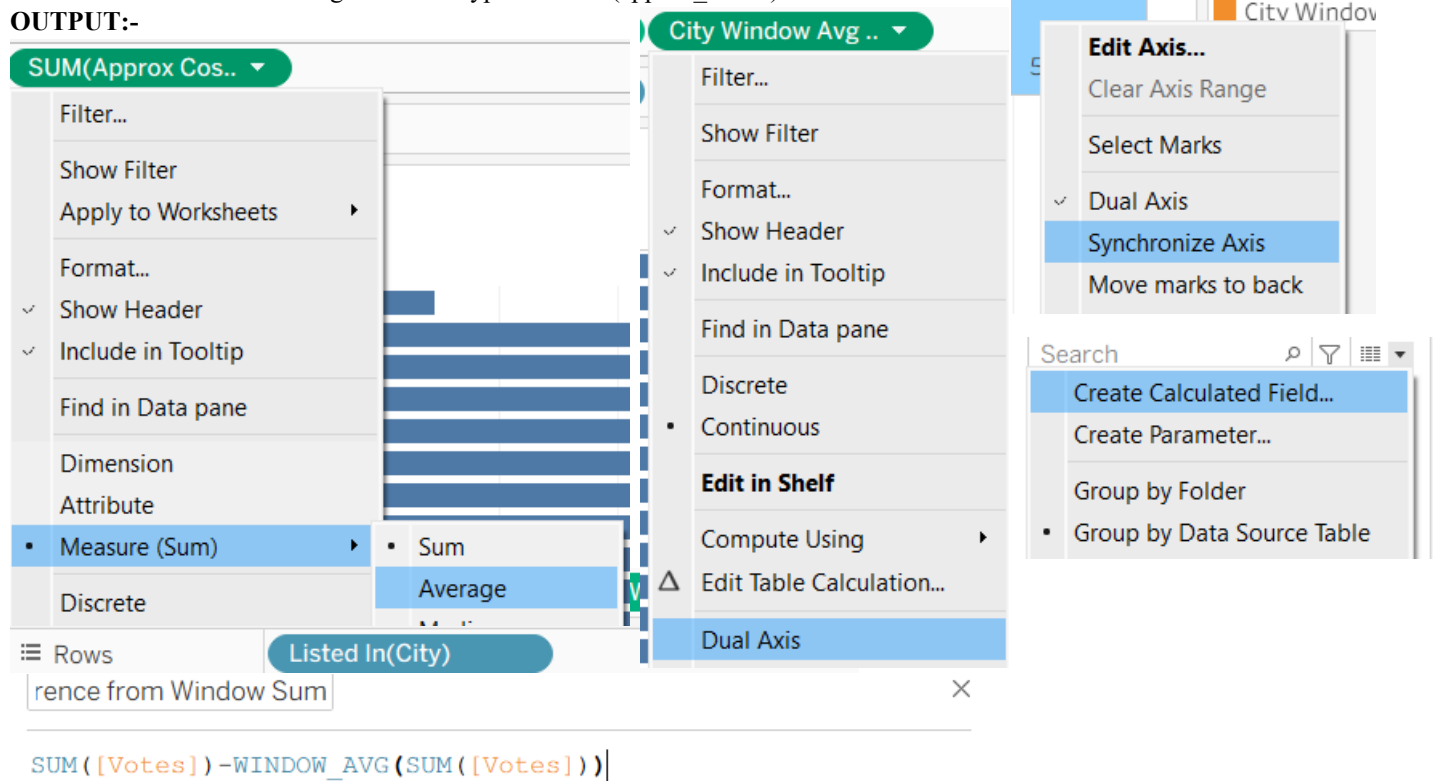


e) Implement window functions (WINDOW_SUM, WINDOW_AVG).

STEPS:-

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.
2. Name the field: City Window Avg Cost.
3. Input the following window function formula: `WINDOW_AVG(AVG([approx_cost(for two people)]))`
4. Open a new worksheet and rename it: Cost vs City Average. Drag listed_in(city) to the Rows shelf.
5. Drag approx_cost(for two people) to the Columns shelf. Change its aggregation to Average.
6. Drag your newly created City Window Avg Cost calculated field onto the Columns shelf
7. Right-click the City Window Avg Cost pill on the Columns shelf and select Dual Axis.
8. Right-click the top horizontal axis in the visualization view and select Synchronize Axis.
9. In the Marks card: Change the mark type for AVG(approx_cost...) to Bar.

OUTPUT:-

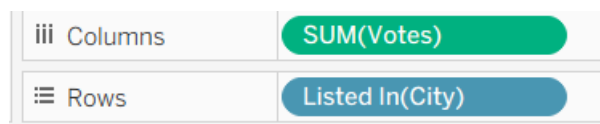
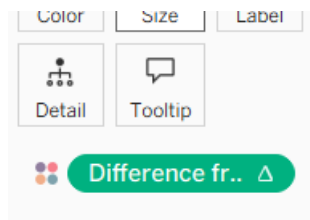


`SUM([Votes]) - WINDOW_AVG(SUM([Votes]))`

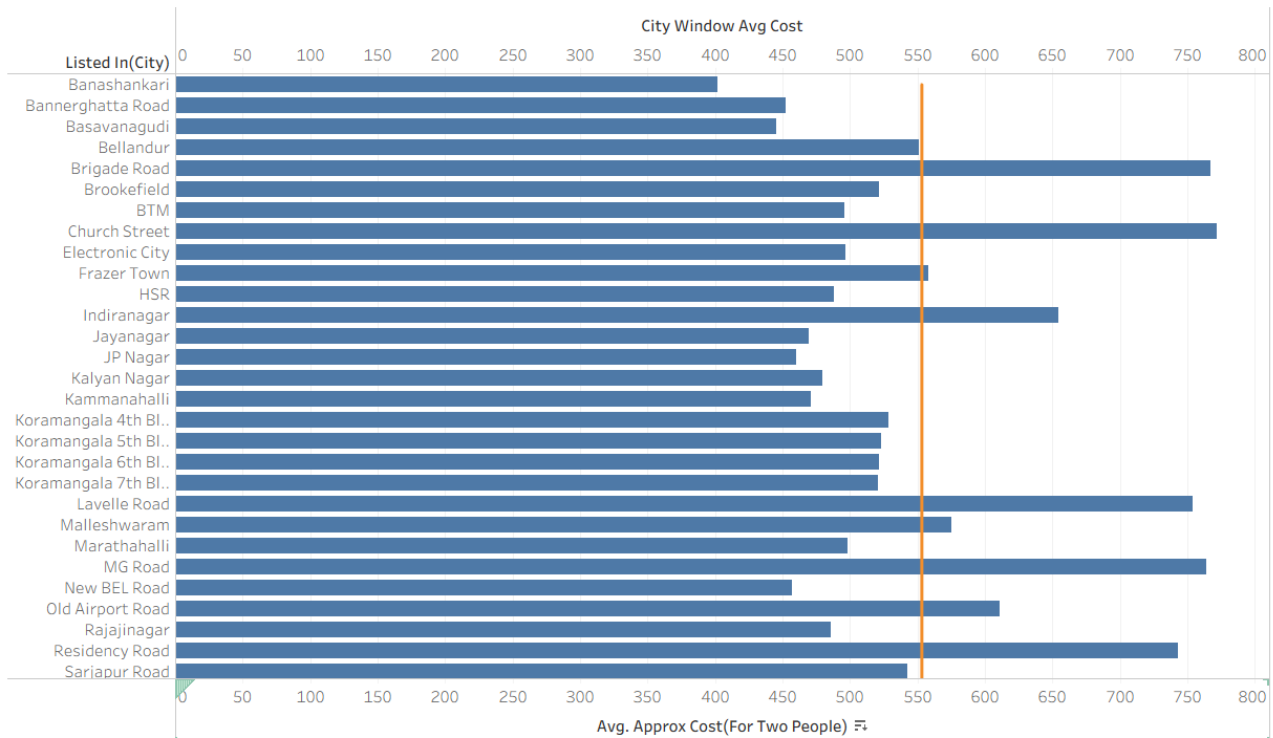
Default Table Calculation

The calculation is valid.

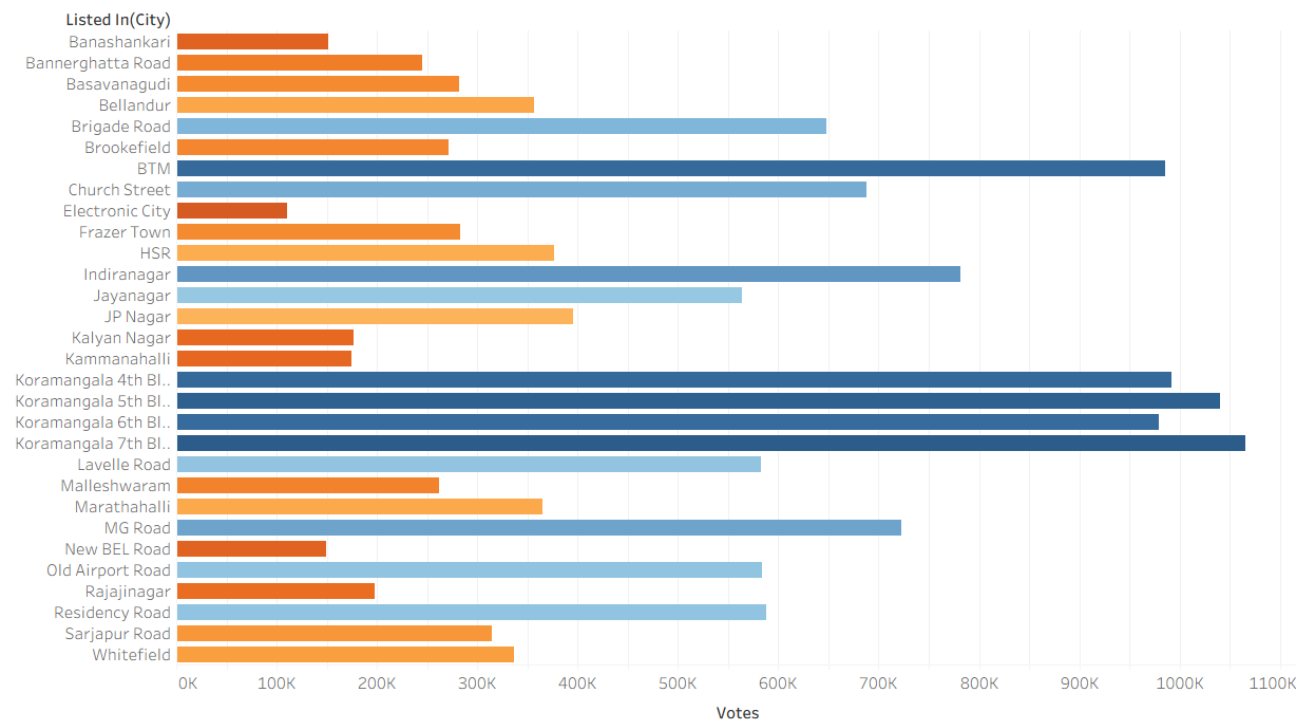
Apply OK



Cost And City Average T054



T054



f) Customize addressing and partitioning.

STEPS:-

1. Open a new worksheet and rename it: Addressing vs Partitioning.
2. Drag listed_in(city) to the Rows shelf.
3. Drag listed_in(type) to the Rows shelf, placing it right next to listed_in(city) (creating a two-level nested hierarchy).
4. Drag votes to the Columns shelf as SUM(votes).
5. Right-click the SUM(votes) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank.
6. Right-click the SUM(votes) pill again and hover over Compute Using: o Select Table (down): Tableau addresses the entire table from top to bottom. It ranks every single row relative to the whole sheet (Partition = Entire Table).
7. Right-click the SUM(votes) pill (with the Δ icon) on the Columns shelf and select Edit Table Calculation...
8. Under Compute Using, select the radio button for Specific Dimensions.
9. Observe the checked vs. unchecked boxes: Checked boxes = Addressing fields: Tableau computes along these dimensions. o Unchecked boxes = Partitioning fields: Tableau restarts the calculation whenever this dimension changes.
10. Uncheck listed_in(city) and leave listed_in(type) checked.
11. Close the Table Calculation dialog box.

OUTPUT:-

Addressing and partitioning

Listed In(City)	Listed In(Type)	Table Calculation
Banashankari	Buffet	Rank of Votes
	Cafes	
	Delivery	

Addressing and partitioning

Listed In(City)

Banashankari

Bannerghatta Road

Basavanagudi

Bellandur

Brigade Road

Listed In(Type)

Buffet

Cafes

Delivery

Desserts

Dine-out

Drinks & nightlife

Buffet

Cafes

Delivery

Desserts

Dine-out

Drinks & nightlife

Buffet

Cafes

Delivery

Desserts

Dine-out

Drinks & nightlife

Pubs and bars

Buffet

Table Calculation

Rank of Votes

Calculation Type

Rank

Descending

Competition (1, 2, 2, 4)

Compute Using

Table (down)

Pane (down)

Pane (across then down)

Pane (down then across)

Cell

Specific Dimensions

☒ Listed In(Type)
 ☐ Listed In(City)

☒ Show calculation assistance

Name

Top N Threshold

Properties

Data type	Display format
Integer ▼	3 ▼

Current value	Value when workbook opens
3	Current value ▼

Allowable values

☐ All
 ☐ List
 ☒ Range

Range of values

☒ Minimum	3	☒ Fixed
☒ Maximum	20	☐ When workbook opens
☒ Step size	1	

Add values from ▼

Cancel

OK

Listed In(City)	Listed In(Type)	Rank of Votes
Banashankari	Buffet	5
	Cafes	3
	Delivery	1
	Desserts	4
	Dine-out	2
Bannerghatta Road	Drinks & nightlife	6
	Buffet	4
	Cafes	5
	Delivery	1
	Desserts	6
Basavanagudi	Dine-out	2
	Drinks & nightlife	3
	Buffet	3
	Cafes	4
	Delivery	1
Bellandur	Desserts	5
	Dine-out	2
	Drinks & nightlife	3
	Pubs and bars	4
	Drinks & nightlife	6
Brigade Road	Buffet	5
	Cafes	7
	Delivery	1
	Desserts	6
	Dine-out	2
Brigade Road	Desserts	7
	Dine-out	1
	Drinks & nightlife	3
	Drinks & nightlife	4
	Drinks & nightlife	6

The image shows the 'Parameters' pane in Tableau Desktop. On the left, a list of parameters is displayed, each with a small icon and a text label. The parameters are: 'Rate' (ABC icon), 'Rest Type' (ABC icon), 'Reviews List' (ABC icon), 'Url' (ABC icon), 'Measure Names' (ABC icon), 'Approx Cost(For Two Pe...' (hash icon), 'City Window Avg Cost' (hash icon), 'Difference from Window ...' (hash icon), 'Restaurant Rank' (hash icon), 'Votes' (hash icon), 'Votes(RUNNING_SUM)' (hash icon), 'Latitude (generated)' (globe icon), 'Longitude (generated)' (globe icon), and 'zomato.csv (Count)' (hash icon). On the right, a context menu is open for the 'Rate' parameter. The menu has a title bar 'Add to Sheet' and a list of actions: 'Show Parameter' (highlighted), 'Cut', 'Copy', 'Edit...', 'Duplicate', 'Rename', 'Hide', 'Delete', 'Create' (with a right arrow), 'Default Properties' (with a right arrow), 'Folders' (with a right arrow), 'Replace References...', and 'Describe...'. The 'Show Parameter' option is selected, and the menu is open for the 'Rate' parameter.

STEPS:-

- OUTPUT:-**

A screenshot of the 'Search' dropdown menu in Tableau. The menu is open, showing two options: 'Create Calculated Field...' and 'Create Parameter...'. The 'Create Calculated Field...' option is highlighted with a blue background.

Dynamic Title

"Top"+STR([Top N Threshold])+"Restaurant by Total Votes"

The calculation is valid.

Apply

OK

Edit Title

Tableau Light

15

B

I

U

Color

Align

Insert

Insert

top and Bottom Restaurants T054

Data Source Name

Data Update Time

Sheet Name

Workbook Name

Page Count

Page Name

Page Number

Default Caption

Default Title

Full Name

User Name

Parameters.Top N Threshold

Reset

OK

Cancel

Edit Title

Tableau Light

15

B

I

U

Color

Align

Insert

top and Bottom Restaurants T054<Parameters.Top N
Threshold>Restaurant by Total Votes

Reset

OK

Cancel

Apply

Phone

Rate

Rest Type

Reviews List

Url

Measure Names

Approx Cost(For Two Pe...

City Window Avg Cost

Difference from Window ...

Restaurant Rank

Votes

Votes(RUNNING_SUM)

Latitude (generated)

Longitude (generated)

Parameters

Top N Threshold

Add to Sheet

Show Parameter

Cut

Copy

Edit...

Duplicate

Rename

Hide

Delete

Create

Default Properties

Folders

Replace References...

Describe...

Size

Label

Tooltip

SUM(Votes)

Magnana Foods

Onesta

Small's Diner Cafe

Show mark labels

Label Appearance

Text: <SUM(Votes)>

Font: Tableau Book, 9pt...

Alignment: Automatic

Marks to Label

All

Selected

Min/Max

Highlighted

Scope

Table

Cell

Pane

Line/Pie

Field

Automatic

Automatic

Name

SUM(Votes)

Label min

Label maximum value

Top N Threshold

8

◀ ▶

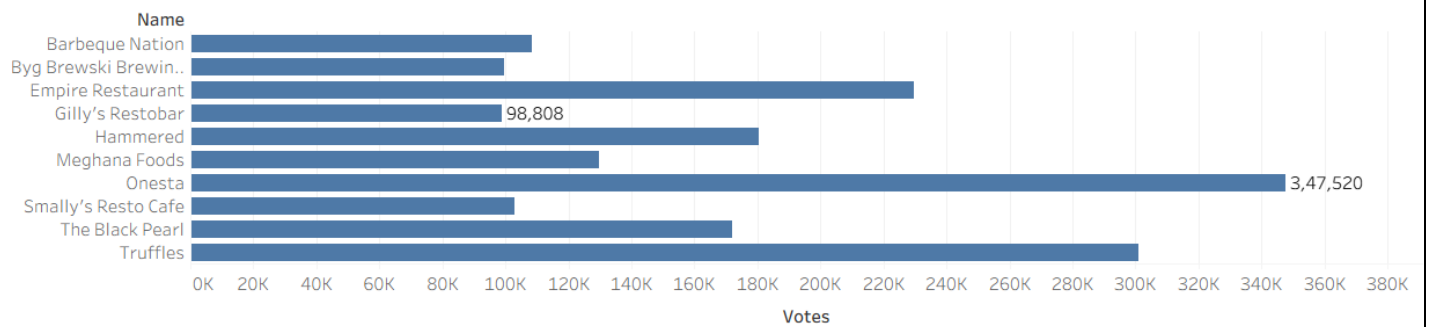
Columns

SUM(Votes)

Rows

Name

top and Bottom Restaurants T0548Restaurant by Total Votes



CONCLUSION:-

The practical was successfully completed using Tableau to analyze data with **table calculations**. We learned how to apply quick table calculations such as running totals and percentages, create moving averages and cumulative metrics, perform year-over-year growth analysis, and rank data using Top N and Bottom N. We also understood the use of window functions, addressing and partitioning, and dynamic titles and labels. Overall, the practical helped in understanding how table calculations can be used to analyze trends, compare data, and gain meaningful insights from datasets.